

Exercise (3.1.41). Let G be a group. Prove that $N = \langle x^{-1}y^{-1}xy | x, y \in G \rangle$ is a normal subgroup of G . and G/N is abelian.

Proof: Let G be a group, and suppose that $N = \langle x^{-1}y^{-1}xy | x, y \in G \rangle$. Consider $g \in G$. Note that for the generator of N , $gx^{-1}y^{-1}xyg^{-1} = gx^{-1}(g^{-1}g)y^{-1}(g^{-1}g)x(g^{-1}g)yg^{-1} = (gx^{-1}g^{-1})(gy^{-1}g^{-1})(gxxg^{-1})(gyyg^{-1}) = (gxg^{-1})^{-1}(gyg^{-1})^{-1}(gxxg^{-1})(gyyg^{-1})$. Let $gxxg^{-1} = a$ and $gyyg^{-1} = b$. Note that $gx^{-1}y^{-1}xyg^{-1} = a^{-1}b^{-1}ab$ therefore by definition $gx^{-1}y^{-1}xyg^{-1} \in N$. Thus $N \trianglelefteq G$.

Since $x^{-1}y^{-1}xy \in N$ for all $x, y \in G$, and $N \trianglelefteq G$ by the previous exercise G/N must be abelian. $\square$

Exercise (3.2.5). Let H be a subgroup of G and fix some element $g \in G$.

a Prove that gHg^{-1} is a subgroup of G of the same order as H .

Proof: Suppose H is a subgroup of G and fix some element $g \in G$. Consider the group gHg^{-1} . Clearly this group is non empty, as it contains the identity. Let $a, b \in gHg^{-1}$ for some $h_a, h_b \in H$ such that $a = gh_ag^{-1}$ and $b = gh_bg^{-1}$. Note that $ab^{-1} = gh_ag^{-1}gh_b^{-1}g^{-1} = gh_ah_b^{-1}g^{-1}$. Since $H \leq G$ we know that $h_ah_b^{-1} \in H$ thus $ab^{-1} \in gHg^{-1}$ and $gHg^{-1} \leq G$.

To show that $|gHg^{-1}| = |H|$ we will prove they are isomorphic. Consider the map $\varphi : H \rightarrow gHg^{-1}$ defined by $\varphi(h) = ghg^{-1}$. Note that this map is a homomorphism, for all $a, b \in H$ we know that $\varphi(ab) = gabg^{-1} = gag^{-1}gbg^{-1} = \varphi(a)\varphi(b)$. This map is also an injection, if $\varphi(a) = \varphi(b)$ then we know that $gag^{-1} = gbg^{-1}$, and thus $a = b$. The map is a surjection by definition. Thus $gHg^{-1} \cong H$ and therefore $|gHg^{-1}| = |H|$. $\square$

b Deduce that if $n \in \mathbb{Z}^+$ and H is the unique subgroup of G of order n then $H \trianglelefteq G$.

Proof: Suppose that H is the unique subgroup of G of order $n \in \mathbb{Z}^+$. Note that by the previous exercise the subgroup gHg^{-1} must have the same order of H , since H has unique order we know $gHg^{-1} = H$. Thus $H \trianglelefteq G$. $\square$